Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building **Python**, **React**, React Native, and TypeScript products. Experienced in search, recommendations, marketplace workflows, caching, automated testing, AI-assisted development, monitoring, and scalable customer-facing systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and maintained a multi-tenant **React** and Python platform for 500+ users, optimizing indexed search paths, filtering, and backend queries to reduce customer-facing latency from 1.2 seconds to 350 milliseconds.
- Built monitored workflows processing 5,000+ monthly jobs with structured outputs, validation, anomaly detection, failure isolation, and recovery paths, sustaining greater than 99% reliability across CI/CD-backed production releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python services handling 10,000+ daily requests, translating user needs into upload, history, filtering, status, and investigation workflows while maintaining stable performance under peak load.
- Used Claude, Codex, and Cursor to accelerate implementation, debugging, tests, and documentation while retaining human review, reducing feedback turnaround 60% and maintaining 90%+ consistency across production workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **ML evaluation pipeline** processing 100,000+ records, improving model quality 18% through reproducible experiments, automated metrics, controlled configurations, performance tracing, and structured analysis of failure modes.
- Collaborated on model evaluation across complex scenarios, improving stability 15% while inspecting outputs, debugging failure patterns, validating robustness, and communicating tradeoffs between learned behavior and expected user-impacting outcomes.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise software with automated builds and cross-platform validation, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito while providing code-review feedback and dependable release support.
- Developed object-oriented services using Python, Java, SQL, caching, filtering, and pagination, improving response times 30% and processing efficiency 68% while strengthening maintainability across high-volume customer-facing application paths.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a two-sided service-booking marketplace with **React Native**, React, TypeScript, Flask, and Firebase, supporting authentication, provider discovery, booking, pricing, payment, status, and notification workflows across mobile and web.
- Mapped 12+ integration points and delivered real-time synchronization, analytics, refunds, and customer-facing updates across 50,000+ records, reducing manual payment handling 70% while maintaining sub-200-millisecond booking responses.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Full Stack: Python, React.js, React Native, Next.js, TypeScript, Flask, FastAPI, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB, Azure SQL
Search & Personalization: Semantic search, FAISS, embeddings, recommendation systems, filtering, pagination, caching, retrieval pipelines, model evaluation, data-driven experimentation
AI Development: Claude Code, Codex, Cursor, LLM APIs, prompt workflows, AI-assisted implementation, debugging, test generation, documentation, human verification
Delivery & Operations: Git, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, AWS, Azure, GCP, Sentry, logging, monitoring, anomaly detection
Engineering Practices: Agile, system design, automated testing, code reviews, production support, cross-functional collaboration, stakeholder communication, technical documentation

Projects

Big-Brain | Live | *Semantic Search, FAISS, PostgreSQL, GCP*
- Built a semantic-search platform processing 50,000+ records with FAISS, PostgreSQL, secure access, and incremental indexing, reducing retrieval latency 40% through optimized embeddings, caching-aware retrieval, and backend pipelines.

Trend Tracker | Live | *React, Recommendations, Material UI, AWS*
- Built a customer-facing recommendation experience serving three million users with React and Material UI, integrating visual personalization and streamlined discovery while improving recommendation accuracy from 78% to 92%.

HMIS | Live | *React, TypeScript, User Workflows, Sentry*
- Built a role-based customer workflow application with scheduling, search, notifications, dashboards, validation, and Sentry monitoring, reducing administrative workload 30% while supporting secure access and reliable post-launch operations.